

DUBUQUE, IA, USA

WMO: 725470

Lat: 42.398N

Lon: 90.704W

Elev: 1056

StdP: 14.14

Time Zone: -6.00 (NAC)

Period: 94-19

WBAN: 94908

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
1	-8.6	-3.2	-15.3	2.5	-7.5	-10.6	3.2	-2.5	27.3	20.0	25.3	22.5	9.0	290	0.579

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	18.1	88.3	75.1	85.6	73.4	83.2	71.6	77.6	85.2	75.6	82.6	73.9	80.6	11.6	200

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	
75.3	137.8	82.3	73.3	128.9	80.6	71.6	121.4	78.4	41.7	85.2	39.8	82.6	38.1	80.6	84.6

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
1%	2.5%	5%	Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
			Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max		
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
25.6	22.6	19.8	-16.3	92.1	6.7	3.0	-21.1	94.3	-25.1	96.1	-28.8	97.8	-33.7	100.0		
			-16.6	80.7	6.2	1.7	-21.1	81.9	-24.7	82.9	-28.2	83.9	-32.7	85.1		
			DB													
			WB													

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
(6) Temperatures, Degree-Days and Degree-Hours	DBAvg	47.6	18.9	23.1	35.4	47.8	58.9	68.6	71.9	70.0	63.0	50.4	36.9	24.7
	DBStd	20.80	12.55	12.08	11.68	9.42	8.28	6.25	5.46	5.17	7.80	9.30	10.14	11.60
	HDD50	3664	965	755	473	148	17	0	0	0	5	109	406	786
	HDD65	7005	1430	1175	918	519	225	35	8	14	129	460	842	1250
	CDD50	2789	0	1	21	82	293	557	680	621	395	123	14	2
	CDD65	653	0	0	1	3	36	142	223	171	69	8	0	0
	CDH74	5095	0	0	6	51	300	1126	1841	1188	517	66	0	0
	CDH80	1352	0	0	0	6	61	305	562	288	119	11	0	0
(14) Wind	WSAvg	9.8	10.9	11.1	11.2	11.9	10.2	8.6	7.3	6.9	8.1	9.8	10.7	10.5
(15) Precipitation	PrecAvg	37.8	1.3	1.4	2.8	3.8	4.3	4.4	3.9	4.6	4.4	2.6	2.7	1.8
	PrecMax	63.4	6.0	3.1	5.9	7.7	9.4	10.5	12.2	9.9	15.5	8.6	10.6	4.1
	PrecMin	22.6	0.3	0.2	0.4	0.8	0.7	0.7	0.9	0.1	0.1	0.0	0.4	0.2
	PrecStd	8.6	1.0	0.8	1.5	1.7	2.0	2.6	2.5	2.7	3.5	1.8	2.0	1.0
(19) Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	49.5	56.8	73.0	80.0	86.1	90.1	93.2	90.6	87.5	81.3	67.8	57.8
		MCWB	43.9	49.3	61.3	62.0	70.2	75.0	77.9	75.4	73.5	67.6	56.5	56.0
	2%	DB	41.8	48.2	64.5	73.7	81.1	86.7	88.6	86.2	83.5	74.4	60.8	49.3
		MCWB	38.5	43.6	55.2	59.4	67.1	73.2	76.6	74.4	70.5	63.1	53.4	46.0
	5%	DB	37.7	42.4	58.1	68.5	77.2	83.7	85.6	83.0	80.1	69.9	56.3	43.3
		MCWB	35.2	38.6	51.4	55.9	64.5	71.1	74.4	72.6	68.0	61.3	50.7	40.2
	10%	DB	34.7	38.1	51.7	63.8	73.4	80.6	83.0	80.4	76.4	65.3	51.8	38.7
		MCWB	32.9	35.3	45.3	53.0	62.7	69.3	72.5	70.9	66.4	57.6	46.4	36.2
(27) Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	44.1	50.8	62.4	65.2	72.8	77.9	81.5	79.0	75.8	70.3	59.2	56.3
		MCDB	46.3	55.2	71.8	75.0	83.0	86.7	89.4	86.9	83.9	77.8	64.9	58.0
	2%	WB	39.0	44.0	57.5	61.4	69.8	75.3	78.3	76.6	73.1	66.3	55.3	46.3
		MCDB	41.9	48.2	62.5	71.0	78.0	83.5	85.8	83.2	80.0	71.5	59.7	48.1
	5%	WB	35.5	39.0	51.7	58.3	67.1	73.4	76.5	74.4	70.8	62.6	51.5	40.4
		MCDB	37.4	42.1	57.8	67.1	74.5	81.3	83.5	80.4	76.9	68.1	55.6	42.7
	10%	WB	33.0	35.5	46.0	54.4	64.5	71.2	74.6	72.6	68.4	58.9	46.7	36.4
		MCDB	34.7	37.8	51.6	62.3	71.3	78.5	80.9	78.4	74.2	64.9	51.2	38.5
(35) Mean Daily Temperature Range	5% DB	MDBR	14.4	14.8	16.7	19.3	19.2	18.2	18.1	17.9	19.8	18.4	15.6	13.5
		MCDBR	16.7	18.0	23.9	26.8	23.1	21.0	20.1	19.8	22.9	23.4	21.7	17.6
	5% WB	MCWBR	13.8	13.9	15.3	15.7	12.5	11.1	10.3	10.4	11.7	13.9	15.6	14.4
		MCWBR	16.0	17.5	22.2	23.6	20.1	18.9	18.1	17.4	19.1	20.2	20.2	16.2
(40) Clear-Sky Solar Irradiance	taub		0.280	0.297	0.333	0.371	0.400	0.419	0.430	0.408	0.379	0.335	0.292	0.278
		taud	2.396	2.354	2.378	2.294	2.273	2.259	2.222	2.285	2.340	2.465	2.512	2.434
	Ebn at Noon		275	288	289	284	278	272	267	269	270	271	271	266
		Edh at Noon	25	30	34	40	42	43	44	40	35	27	22	22
(44) All-Sky Solar Radiation	RadAvg		562	880	1191	1469	1735	1934	1933	1697	1413	918	608	444
	RadStd		66	95	119	119	122	158	124	114	112	114	47	39

Historical Trends

	DBAvg	Heating		Cooling			Degree-Days			
		99% DB	99% DP	1% DB	1% WB	1% DP	HDD50	HDD65	CDD50	CDD65
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)
(46) Station Only		N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
(47) Regional (46 neighbors)		N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A

Nomenclature: See separate page